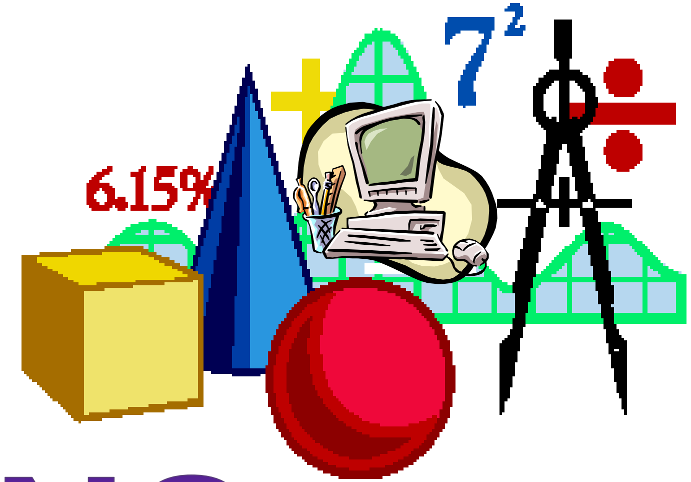
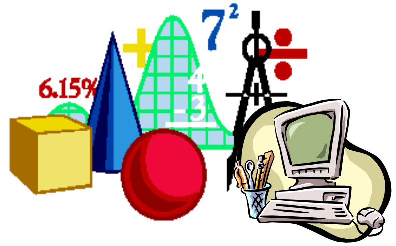
Introduction
to



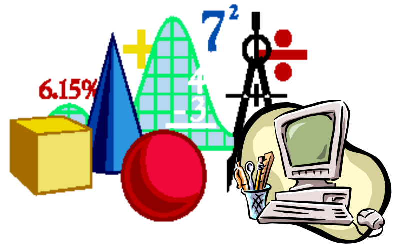
OPERATIONS RESEARCH



OPERATIONS RESEARCH

- The **Operations Research** (**OR**) has come to solve, in an efficient way, problems that involve the optimal management of limited resources

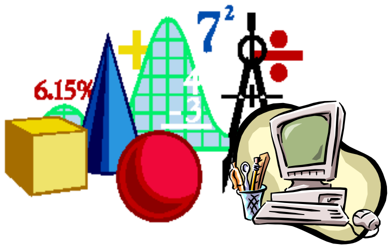
- Its origin as a science dates from 1947 when George Dantzig invented the "simplex method" for solving optimization problems formulated from logistical issues of the United States Air Force during the Second World War



OPERATIONS RESEARCH

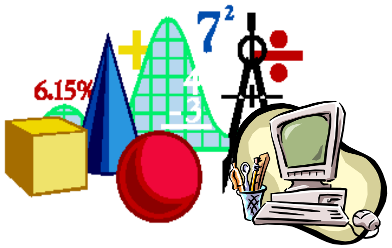
- Applies a **quantitative analysis** to real complex problems that involve a **decision taking**, using a set of methods that are essentially based on **mathematical procedures**

- The goal is to find the **best solution** for these problems, that is, the **optimal solution**, in order to provide the best decision



LINEAR PROGRAMMING

- **Linear Programming** (**LP**) is one of the most developed and most utilized fields of OR
- It optimizes decision problems by representing them in terms of a **mathematical model of LP**
- The LP model is characterized by the fact that all **mathematical expressions** that compose it are **linear**



LINEAR PROGRAMMING

Formulating a problem in terms of a LP model consists in specifying:

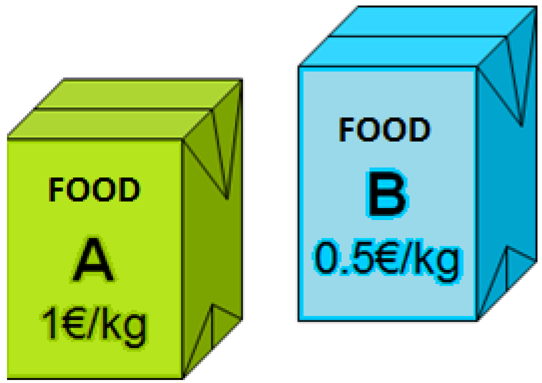
- **Decision variables** (what we want to determine)
- **Objective function** (what we want to optimize)
- **Constraints** (conditions that should be satisfied)

EXAMPLE OF A LP PROBLEM

The dilemma of Mr. Joshua

O Mr. Joshua is dedicated to the breeding and sale of dogs of a particular breed with a high market demand. Since he wants his animals grow healthy and beautiful, he knows he must give them a balanced diet.

In fact, Mr. Joshua has at its disposal two types of food, **A** and **B**, with different features and prices.



Composition in terms of nutrients of food **A** e **B**:

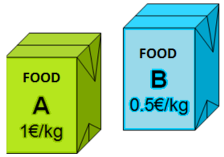
	Food	
Nutrients	A (g/kg)	B (g/kg)
Minerals	20	50
Vitamins	50	10
Calcium	30	30

Minimal amounts of nutrients, each week, for a balanced diet (according to veterinarians):

Nutrients	Minimum amount required (in g)
Minerals	200
Vitamins	150
Calcium	210

Mr. Joshua thinks about what he wants:

- On the one hand he wants to **follow the instructions given by veterinarians** in order to provide an adequate diet to his dogs



- But on the other hand, he wants to minimize spending on animal feed

So, the question to solve is the following:

Which amount of each type of food (**A** and **B**) should Mr. Joshua give to each dog per week in order to:

- Respect the recommended minimum amounts of food and
- Minimize the cost of feeding each dog

To determine the answer to this question, it is necessary to translate the problem into a **mathematical model of LP**

Linear programming model

- **Decision variables:**

x_1 – Amount (in Kg) of food of type **A** to give each dog per week

x_2 – Amount (in Kg) of food of type **B** to give each dog per week

- **Objetive function:**

minimize cost, that is,

$$\min z = 1\, x_1 + 0.5\, x_2$$

- **Contraints:**

$$20\,x_1 + 50\,x_2 \geq 200 \quad \Big\} \text{ Minerals}$$

$$50\,x_1 + 10\,x_2 \geq 150 \quad \Big\} \text{ Vitamins}$$

$$30\,x_1 + 30\,x_2 \geq 210 \quad \Big\} \text{ Calcium}$$

$$x_1 \geq 0,\ x_2 \geq 0$$

Determination of the optimal solution

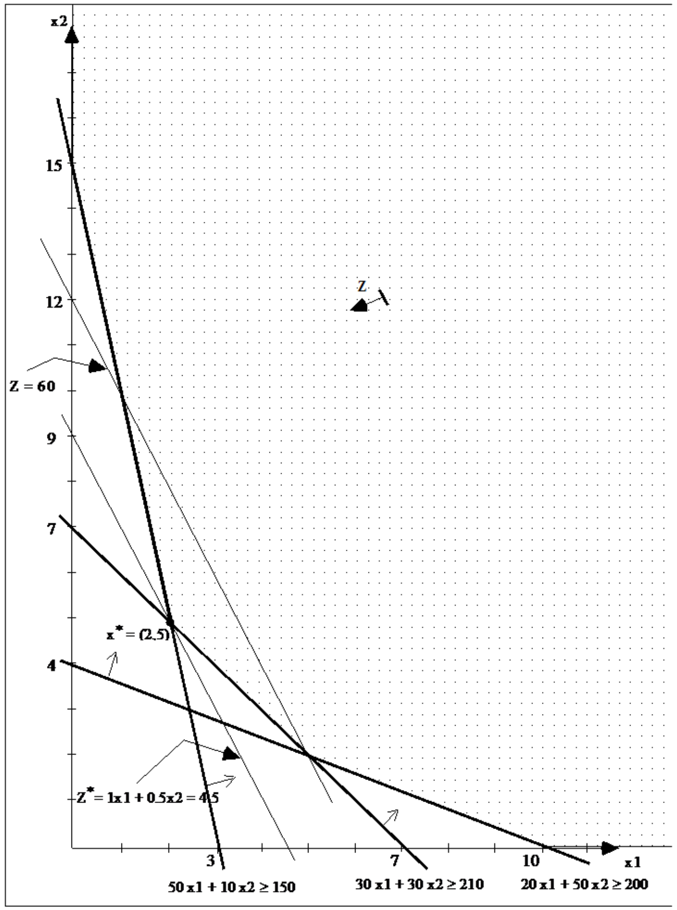
- **Graphical method**

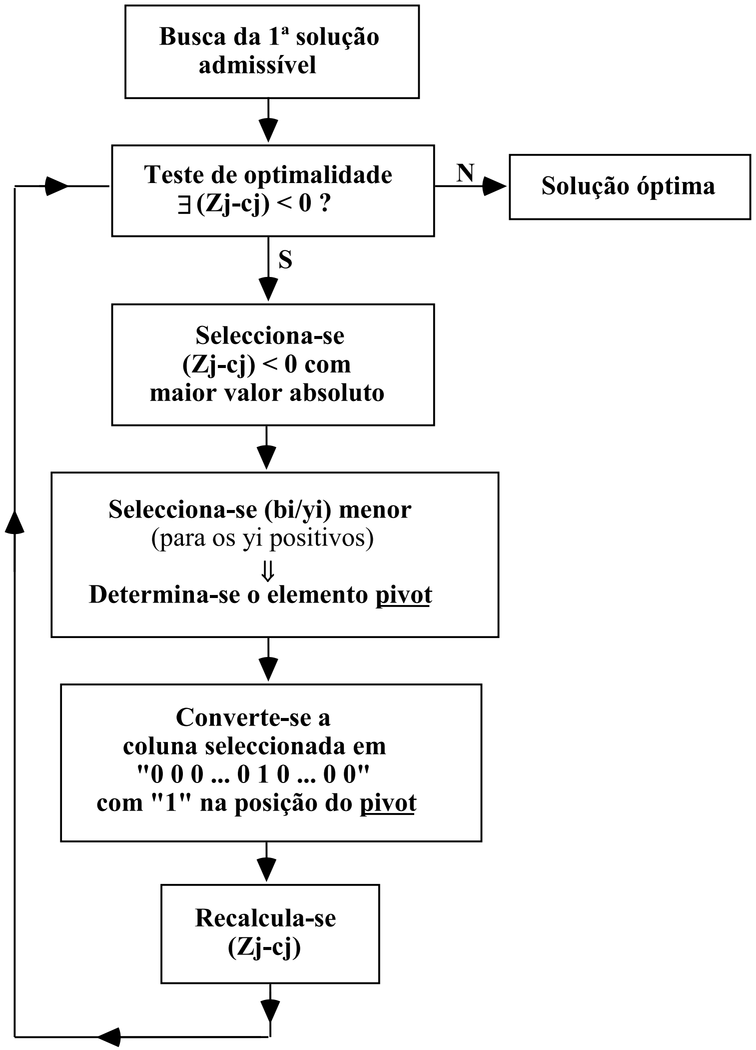
  Used to solve simple problems

- **Algebraic method**

  One of the linear programming algorithms

Graphical method / Algebraic method





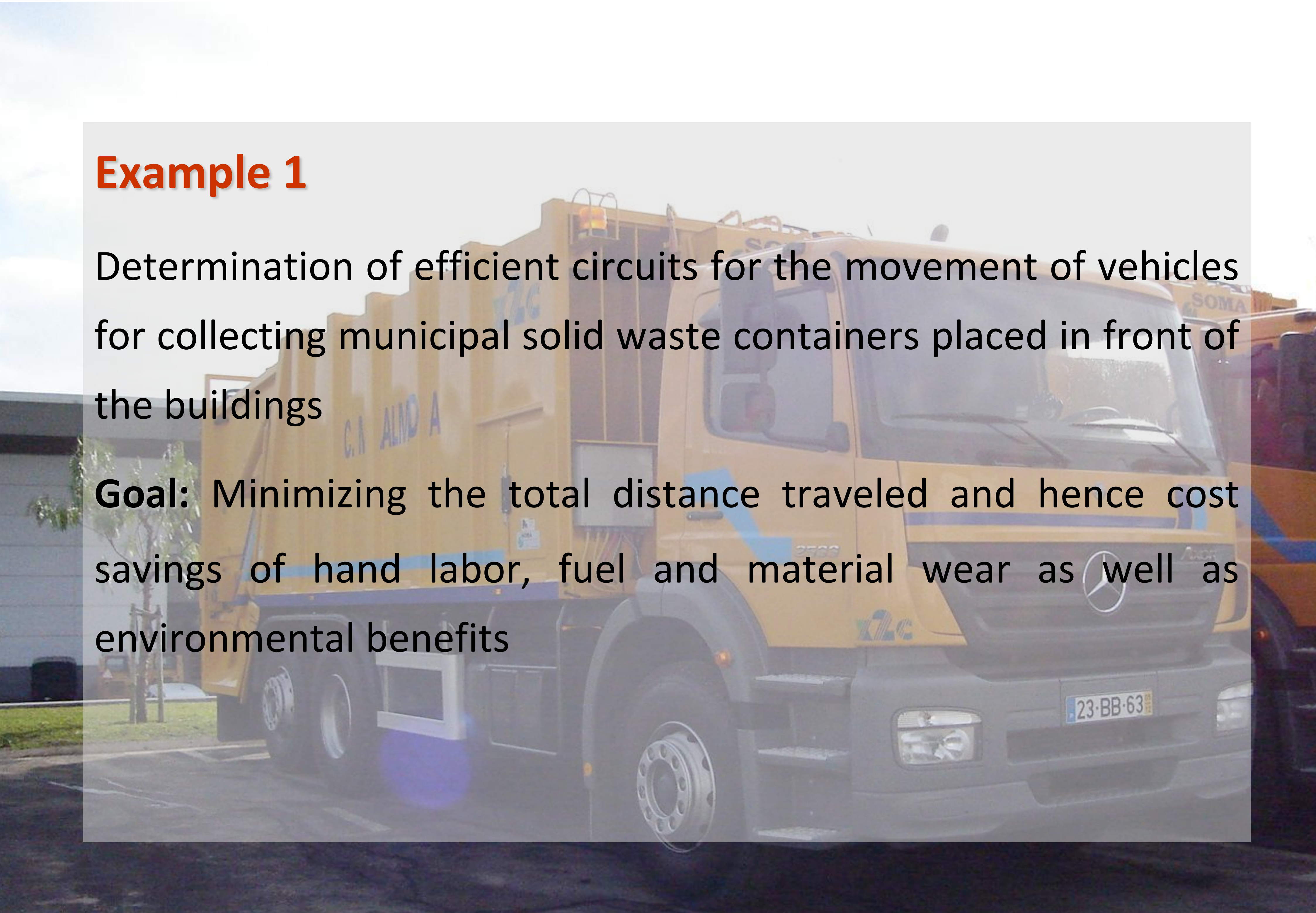
Solution presentation

Solving by one of the methods previously referred, we would obtain $\mathbf{x_1=2}$, $\mathbf{x_2=5}$ e $\mathbf{z=4.5}$. Thus, the solution to present to Mr. Joshua would be the following:

He should feed each dog with **2 kg** of food type **A** and **5 kg** of food type **B**, per week, in order to give animals the nutrients that are essential to a healthy growth, spending a minimum of **4.5 €**, per week..

Examples of real applications

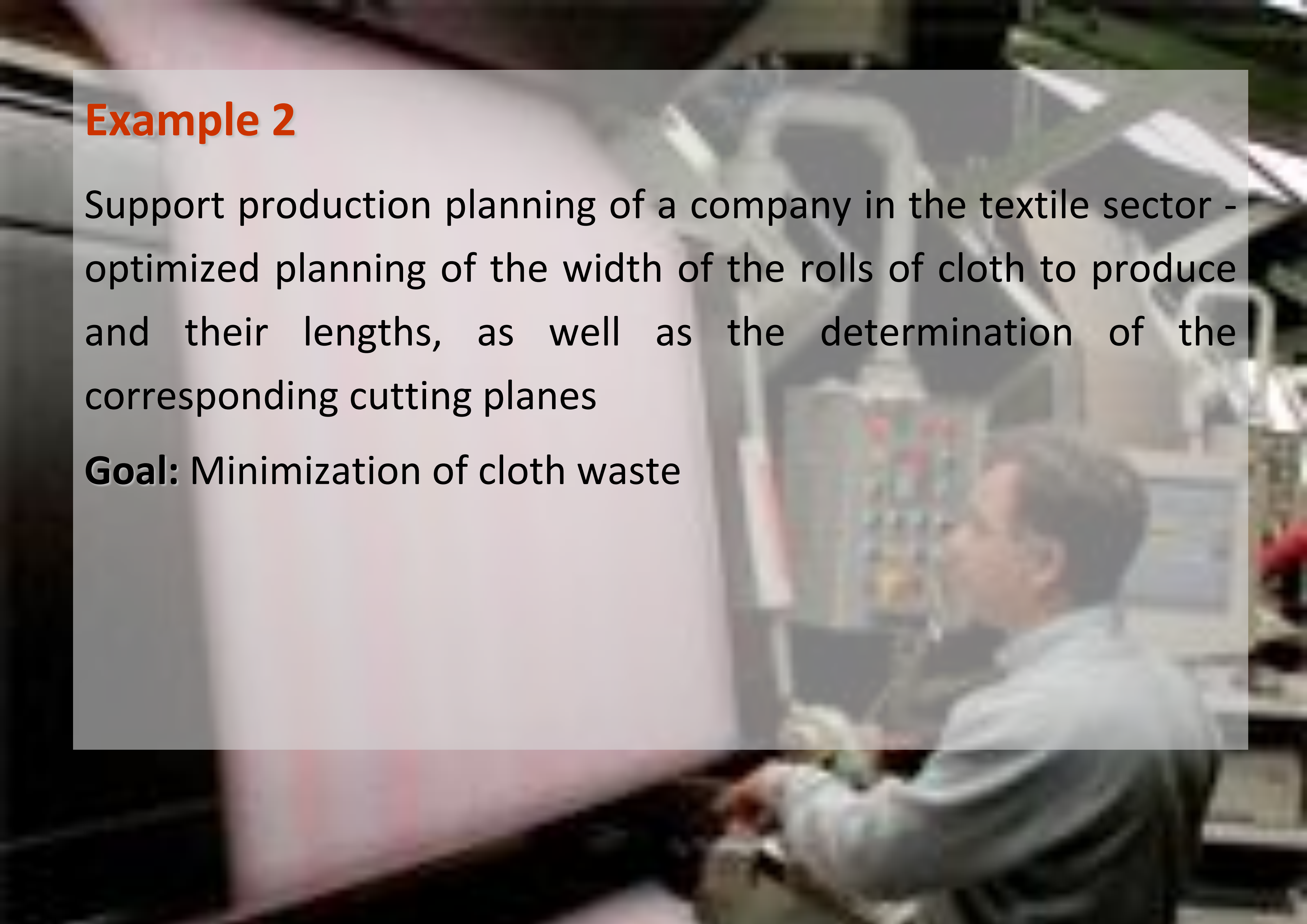
The examples were retrieved from “Casos de Aplicação da Investigação Operacional”, C. H. Antunes e L. V. Tavares, McGraw-Hill Portugal, 2000



Example 1

Determination of efficient circuits for the movement of vehicles for collecting municipal solid waste containers placed in front of the buildings

Goal: Minimizing the total distance traveled and hence cost savings of hand labor, fuel and material wear as well as environmental benefits



Example 2

Support production planning of a company in the textile sector - optimized planning of the width of the rolls of cloth to produce and their lengths, as well as the determination of the corresponding cutting planes

Goal: Minimization of cloth waste